Sean Luka Girgis

Dynatrace / APM Observability Engineer | RUM, Synthetic Monitoring & Telemetry

214-315-2190 | seanlgirgis@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Senior APM, observability, performance, and capacity engineering professional with 20+ years of enterprise technology experience across Dynatrace AppMon, Gomez Synthetic Monitoring, CA APM/Introscope, AppDynamics, BMC TrueSight, infrastructure telemetry, dashboards, alerting, synthetic monitoring, RUM/EUM concepts, performance analysis, documentation, and production troubleshooting. Strong background supporting enterprise monitoring platforms, onboarding application and infrastructure telemetry, building dashboards and reports, defining thresholds, reducing troubleshooting friction, and translating telemetry into actionable operational insights. Best aligned to Dynatrace/APM observability roles requiring monitoring administration, application visibility, synthetic monitoring, dashboards, reporting, performance analysis, Python/shell automation, documentation, and knowledge sharing; Terraform, Ansible, PowerShell, and Dynatrace API automation are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Managed and integrated enterprise telemetry from BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to support infrastructure monitoring, capacity analytics, and operational decision-making.
- Built Python/Pandas/PySpark pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw performance signals into validated dashboards, reports, forecasting datasets, and executive insights.
- Supported monitoring and incident-reduction efforts by identifying underutilized resources, bottleneck risks, abnormal trends, capacity pressure, and data quality issues across large banking infrastructure.
- Developed KPI-based reporting and dashboards that converted utilization, application performance, and infrastructure telemetry into actionable views for technical teams and senior stakeholders.
- Collaborated with infrastructure, application, analytics, architecture, and business teams to translate monitoring signals into practical recommendations for operational stability and service delivery.
- Created runbooks, metric definitions, validation checkpoints, reporting logic, support procedures, and knowledge-sharing documentation to improve repeatability and team onboarding.
- Troubleshot production monitoring and reporting issues by tracing source feeds, transformation logic, metric behavior, latency symptoms, failed outputs, and downstream reporting discrepancies.
- Used forecasting workflows with Prophet and scikit-learn to support proactive capacity planning, bottleneck prediction, and risk mitigation for enterprise service delivery.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, supporting application performance visibility and operational readiness.

- Supported application and infrastructure monitoring by configuring dashboards, analyzing synthetic monitoring behavior, and correlating user-facing symptoms with application performance signals.
- Led the FAST project, mining real-user and synthetic monitoring metrics to surface optimization opportunities adopted by engineering teams.
- Supported End-User Monitoring concepts, transaction tracing, dashboarding, and performance reporting for web and mobile experiences.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior with APM telemetry and improve issue diagnosis.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Led and coordinated large-scale CA APM upgrades, monitoring standards, dashboard rollouts, threshold design, alerting practices, and operational reporting adoption across technical teams.
- Designed custom Management Modules, dashboards, alerts, SLA thresholds, and documentation standards used by application, middleware, infrastructure, and operations teams.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual performance-reporting workflows.
- Provided technical training, onboarding support, troubleshooting guidance, and documentation to improve adoption of monitoring standards and operational playbooks.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, regression thresholds, test findings, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Supported performance-focused cutover planning, validation, and troubleshooting for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Flagship Projects

FAST Performance Analytics Project

Mined Dynatrace and Gomez Synthetic Monitoring data to identify optimization opportunities, improve transaction visibility, and communicate actionable findings to engineering teams.

Technologies: Dynatrace AppMon, Gomez Synthetic Monitoring, Performance Analytics, Dashboards, APM

Enterprise APM Reporting Automation

Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry across large CA APM enterprise monitoring environments.

Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Monitoring, Automation

HorizonScale — Capacity Forecasting Engine

Built Python/PySpark forecasting workflows to process infrastructure telemetry, prepare forecasting datasets, validate pipeline outputs, and deliver capacity and bottleneck insights.

Technologies: Python, PySpark, Prophet, scikit-learn, BMC TrueSight, Telemetry